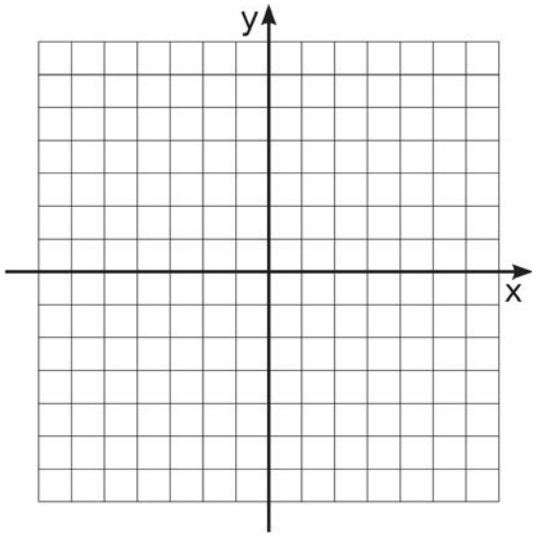
1. Find the volume of the region bounded by $y = \sqrt{x}$, $y = 0$, and $x = 1$ rotated around the x-axis by following the following steps. *Do not rush to part (d). Spend time on (a) to (c).*

   (a) Draw the region (make it big). Include a line segment, ℓ, that when rotated around the axis or revolution indicates the disc method is to be used. Label the graph.



   (b) If extended, the line segment ℓ would intercept the _____-axis. Thus, the variable of integration is _____.

   (c) The line segment ℓ is _____ to the axis of rotation.

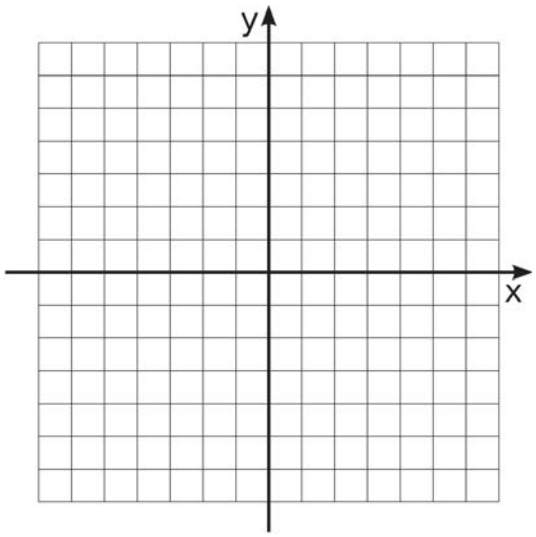
   (d) Set up an integral for the volume of the solid.

   (e) Solve the integral. Does your answer seem reasonable?

   (f) Which of the 3-D printed solids (A, B, or C) is described here?

2. Find the volume of the region bounded by $y = \sqrt{x}$, $y = 1$, and $x = 0$ rotated around the x-axis by following the following steps.

   (a) Draw the region (make it big). Include a line segment, ℓ, that when rotated around the axis or revolution indicates the disc method is to be used. Label the graph.



   (b) If extended, the line segment ℓ would intercept the _____-axis. Thus, the variable of integration is _____.

   (c) The line segment ℓ is _____ to the axis of rotation.

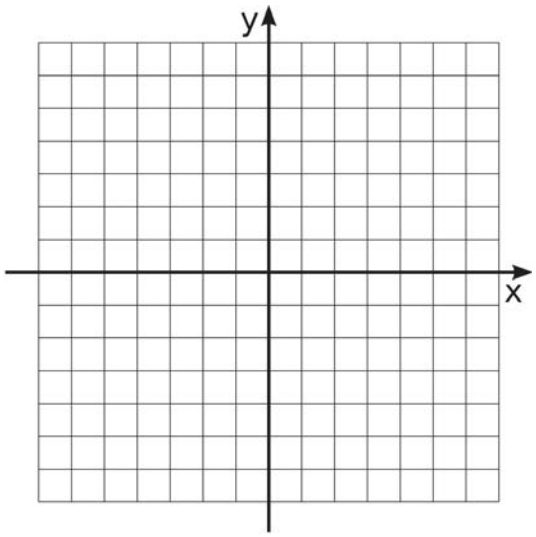
   (d) Set up an integral for the volume of the solid.

   (e) Solve the integral. Does your answer seem reasonable?

   (f) Which of the 3-D printed solids (A, B, or C) is described here?

3. Find the volume of the region bounded by $y = \sqrt{x}$, $y = 1$, and $x = 0$ rotated around the y-axis by following the following steps.

(a) Draw the region (make it big). Include a line segment, ℓ, that when rotated around the axis or revolution indicates the disc method is to be used. Label the graph.



(b) If extended, the line segment ℓ would intercept the _____-axis. Thus, the variable of integration is _____.

(c) The line segment ℓ is _____ to the axis of rotation.

(d) Set up an integral for the volume of the solid.

(e) Solve the integral. Does your answer seem reasonable?

(f) Which of the 3-D printed solids (A, B, or C) is described here?